

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1502

Carrizo Plains 100 kV
January 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1502. The Interconnection Customer proposes a 188.7 MW large load (data center) facility interconnecting at the Carrizo Plains 100 kV substation in the Illinois area. The request was received on June 23, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 459 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,117,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Spills-Carrizo Plains 100 kV
Contingency: Carrizo Plains-N Knox 100 kV Line
Loading: 139.6% (Rating: 122 MVA)
DFAX: 75.0%
Pre-project loading: 23.1%
2. Facility: Carrizo Plains Substation-North Knox 100 kV L
Contingency: Basscreek Substation-Wallisville 100 kV Line
Loading: 138.3% (Rating: 137 MVA)
DFAX: 41.9%
Pre-project loading: 80.6%
3. Facility: Basscreek-Wallisville 100 kV
Contingency: North Knox-Carrizo Plains 100 kV L
Loading: 119.9% (Rating: 139 MVA)
DFAX: 25.0%
Pre-project loading: 86.0%
4. Facility: N Knox-Stuttgart Ricuskey 100 kV Ckt 1
Contingency: Stuttgart Ricuskey-North Knox Substation 100 kV Circuit 2
Loading: 110.1% (Rating: 140 MVA)
DFAX: 31.6%
Pre-project loading: 67.6%
5. Facility: North Knox-Stuttgart Ricuskey 100 kV Cir. 2
Contingency: Stuttgart Ricuskey-North Knox 100 kV Ckt 1
Loading: 110.1% (Rating: 140 MVA)
DFAX: 31.6%
Pre-project loading: 67.6%

6. Facility: Wallisville Substation-Lullwater 100 kV Ckt 2
Contingency: Wallisville-Lullwater 100 kV #1
Loading: 108.8% (Rating: 158 MVA)
DFAX: 8.7%
Pre-project loading: 98.4%
7. Facility: Battle Mtn Substation-Spills 100 kV Ckt 1
Contingency: Spills-Battle Mountain 100 kV
Loading: 108.8% (Rating: 140 MVA)
DFAX: 24.4%
Pre-project loading: 75.9%
8. Facility: Battle Mountain-Spills 100 kV line
Contingency: Battle Mtn-Spills 100 kV Cir. 1
Loading: 108.8% (Rating: 140 MVA)
DFAX: 24.4%
Pre-project loading: 75.9%
9. Facility: Dairymans-Pueblo Xtra 230 kV #2
Contingency: Dairymans-Pueblo Xtra 230 kV Ckt 1
Loading: 107.9% (Rating: 1,013 MVA)
DFAX: 9.0%
Pre-project loading: 106.2%
10. Facility: Dairymans-Pueblo Xtra 230 kV Line
Contingency: Pueblo Xtra-Dairyman 230 kV Line 2
Loading: 106.4% (Rating: 1,027 MVA)
DFAX: 9.0%
Pre-project loading: 104.8%
11. Facility: Henry D King-Basscreek 100 kV
Contingency: Carrizo Plains-North Knox 100 kV
Loading: 106.2% (Rating: 135 MVA)
DFAX: 25.0%
Pre-project loading: 71.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Spills-Carrizo Plains 100 kV	Rebuild line with higher-capacity conductor	\$2,414,000	96% (\$2,313,000)
North Knox-Carrizo Plains 100 kV line	Rebuild line with higher-capacity conductor	\$4,114,000	75% (\$3,079,000)
Basscreek-Wallisville 100 kV	Rebuild line with higher-capacity conductor	\$6,497,000	88% (\$5,740,000)
N Knox-Stuttgart Ricuskey 100 kV Ckt 1	Rebuild line with higher-capacity conductor	\$2,396,000	74% (\$1,776,000)
N Knox-Stuttgart Ricuskey 100 kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,290,000	74% (\$956,000)
Wallisville-Lullwater 100 kV Cir. 2	Reconductor line	\$2,100,000	37% (\$776,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Battle Mtn Substation-Spills 100 kV Ckt 1	Reconductor line	\$1,631,000	58% (\$949,000)
Battle Mountain-Spills 100 kV #2	Reconductor line	\$1,035,000	58% (\$602,000)
Dairymans-Pueblo Xtra 230 kV #2	Reconductor line	\$25,418,000	1% (\$330,000)
Dairymans-Pueblo Xtra 230 kV	Reconductor line	\$16,888,000	1% (\$219,000)
Henry D King-Basscreek 100 kV	Reconductor line	\$426,000	88% (\$377,000)
	Total allocated to project		\$17,117,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.